NCERT • GRADE 7 • MATHS • PART 2

Finding Common Ground

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Prime factorisation

HCF • LCM • Divisibility

Six NCERT *Figure it Out* sets explained

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. All illustrations are editable LaTeX drawings.

Part A • NCERT questions with worked solutions

The two most important rules

HCF (highest common factor): multiply common prime factors using the *smallest* power.
LCM (lowest common multiple): multiply every prime needed using the *largest* power.
We include all six *Figure it Out* sets in textbook order, with page references.

Set 1 • p. 51: List every factor

(a) 90: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90.
(b) 105: 1, 3, 5, 7, 15, 21, 35, 105.
(c) 132: 1, 2, 3, 4, 6, 11, 12, 22, 33, 44, 66, 132.
(d) 360 (24 factors): 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30, 36, 40, 45, 60, 72, 90, 120, 180, 360.
(e) 840 (32 factors): 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 15, 20, 21, 24, 28, 30, 35, 40, 42, 56, 60, 70, 84, 105, 120, 140, 168, 210, 280, 420, 840.

How to check: $360 = 2^3 \times 3^2 \times 5$ has $(3+1)(2+1)(1+1) = 24$ factors. $840 = 2^3 \times 3 \times 5 \times 7$ has $4 \times 2 \times 2 \times 2 = 32$ factors.

Set 2 • p. 53: Common factors and HCF

Numbers	Common factors	HCF
50, 60	1, 2, 5, 10	10
140, 275	1, 5	5
77, 725	1	1
370, 592	1, 2, 37, 74	74
81, 243	1, 3, 9, 27, 81	81

One worked example: $370 = 2 \times 5 \times 37$ and $592 = 2^4 \times 37$. Hence the common prime part is $2 \times 37 = \boxed{74}$.

Set 3 • p. 54: HCF using prime factors

Numbers	Common prime factors	HCF
24, 180	$2^2 \times 3$	12
42, 75, 24	3	3
240, 378	2×3	6
400, 2500	$2^2 \times 5^2$	100
300, 800	$2^2 \times 5^2$	100

Q2: If $72 = 6 \times 12$ and $144 = 8 \times 18$, this *does not* prove the numbers have HCF 1. Composite factors can hide shared prime factors. Since $144 = 2 \times 72$, $\mathrm{HCF}(72, 144) = 72$.

Set 4 • p. 58: LCM using prime factors

Numbers	LCM prime factorisation	LCM
30, 72	$2^3 \times 3^2 \times 5$	360
36, 54	$2^2 \times 3^3$	108
105, 195, 65	$3 \times 5 \times 7 \times 13$	1365
222, 370	$2 \times 3 \times 5 \times 37$	1110

Example: $30 = 2 \times 3 \times 5$ and $72 = 2^3 \times 3^2$, so LCM $= 2^3 \times 3^2 \times 5 = 360$.

Set 5 • p. 59: Generalise HCF and LCM

1. HCF patterns:

(a) Consecutive even numbers $2n, 2n+2$: **HCF** $= 2$ because $n, n+1$ are consecutive.

(b) Consecutive odd numbers: **HCF** $= 1$; their difference is 2 and both are odd.

(c) Two even numbers: their HCF is **at least 2**, but it can be greater (e.g., HCF of 12 and 18 is 6).

(d) Consecutive natural numbers: **HCF** $= 1$.

(e) Co-prime numbers: **HCF** $= 1$ by definition.

2. LCM equals one number when one number is a multiple of the other, e.g., $\text{LCM}(4, 20) = 20$. In general $\text{LCM}(a, ka) = ka$ for a positive integer k.

3. LCM patterns:

(a) Two multiples of 3: LCM is **a multiple of 3** (not necessarily just 3).

(b) Consecutive evens $2n, 2n+2$: LCM $= \boxed{2n(n+1)}$.

(c) Consecutive positive integers $n, n+1$: LCM $= \boxed{n(n+1)}$.

(d) Two co-prime numbers a, b: LCM $= \boxed{ab}$.

Set 6 • pp. 63–64: Q1–Q3

Q1. Repeating blue stars: Top-row blue occurs at positions $4, 10, 16, \ldots$ (every 6). Bottom-row blue occurs at $4, 8, 12, 16, \ldots$ (every 4). The next common position after 4 is $4 + \text{LCM}(6, 4) = \boxed{16}$.

Q2. Factor or multiple? Let $A = 5 \cdot 7 \cdot 11^2$ and $B = 5 \cdot 7^2 \cdot 11 \cdot 2$.

(a) A is **not a multiple** of B: A lacks a factor 2 and a second 7.

(b) A is **not a factor** of B: B lacks a second 11.

Q3. Prime forms:

- (a) $A = 3^2 \cdot 5 \cdot 7^2$, $B = 12 \cdot 7 \cdot 11 = 2^2 \cdot 3 \cdot 7 \cdot 11$.
  HCF $= \boxed{3 \cdot 7 = 21}$; LCM $= \boxed{2^2 \cdot 3^2 \cdot 5 \cdot 7^2 \cdot 11 = 97020}$.
- (b) $45 = 3^2 \cdot 5$, $36 = 2^2 \cdot 3^2$.
  HCF $= \boxed{3^2 = 9}$; LCM $= \boxed{2^2 \cdot 3^2 \cdot 5 = 180}$.

Set 6 • pp. 63–64: Q4–Q8

Q4. One valid pair with HCF 1 and LCM 66 is $\boxed{2, 33}$: they are co-prime and their product is 66.

Q5. Cowherd, 3/5/7 gates: Number is a multiple of $\text{LCM}(3, 5, 7) = 105$. Less than 200, hence $\boxed{105}$ **cows**.

Q6. Cubes fitting exactly in a $12 \times 18 \times 36$ cm box: Cube edge divides all three dimensions. HCF $= 6$. From choices 9, 6, 4, 3, 2 cm, **6, 3, 2 cm** work (options b, d, e).

Q7. Largest listed divisor of 306 and 36: $\text{HCF}(306, 36) = \boxed{18}$ (option c).

Q8. A multiple of 3,4,5,7 is $420k$. We need remainder 10 on division by 11. Since $420 \equiv 2 \pmod{11}$, $2k \equiv 10 \pmod{11}$. Smallest $k = 5$ gives $\boxed{2100}$.

Set 6 • pp. 63–64: Q9–Q10

Q9. Children divisible into groups of 6 and 9 but not 10: among the choices, **72 and 36** work (a, e). Both are divisible by 18 and not by 10.

Q10. Two *different* prime numbers m, n are co-prime; $\text{LCM}(m, n) = mn$, which is greater than each. **Only (c) is correct.**

Set 6 • pp. 63–64: Q11

Q11. The rabbit begins 150 ft ahead. Each leap closes the gap by $9 - 7 = 2$ ft. Number of simultaneous leaps: $150/2 = \boxed{75}$.

Set 6 • pp. 63–64: Q12–Q13

Q12. $\mathrm{LCM}(1, 2, 3, 4, 5, 6, 8, 9, 10) = 2^3 \cdot 3^2 \cdot 5 = \boxed{360}$.
Q13. Mahaviracharya's fractions:

$$\frac{8}{15} + \frac{1}{20} + \frac{7}{36} + \frac{11}{63} + \frac{1}{21} = \boxed{1}.$$

A common denominator is 1260: $672 + 63 + 245 + 220 + 60 = 1260$. **Check:** numerator equals denominator, so sum is 1.

Part B • Important textbook explorations

Tiling a 12×16 ft room

Largest integer-square tile: $\mathrm{HCF}(12, 16) = 4$ ft per side. Tiles needed $= (12/4)(16/4) = 3 \cdot 4 = \boxed{12}$. In a regular square grid, the largest exact-tiling side is still 4 ft: both $12/s$ and $16/s$ must be whole-number counts.

84 kg and 108 kg of rice into equal full bags

$\mathrm{HCF}(84, 108) = 12$ kg, so use $84/12 = 7$ and $108/12 = 9$ bags, $\boxed{16 \text{ bags}}$ total.

Jump Jackpot • longest jump size

(14,30) → 2; (7,11) → 1; (30,50) → 10; (28,42) → 14. Each longest jump is the **HCF** of the two numbers.

The toran, Monday gajak and Idli–Vada puzzles

Strips of 6 and 8 cm: shortest shared toran is LCM= $\boxed{24 \text{ cm}}$. A shop gives sweets weekly; visits are every 10 days: meet next after $\mathrm{LCM}(7, 10) = \boxed{70 \text{ days}}$.
Idli–Vada first shared calls: (4,6)→ 12; (7,11)→ 77; (14,30)→ 210; (15,55)→ 165.

Explore factors, scaling and the shortcut

Factors of 840: $28, 14, 8$ all divide 840; 27 does not.
Two multiples of 14: $\mathrm{HCF}(14 \cdot 6, 14 \cdot 9) = 14 \cdot 3 = \boxed{42}$, not simply 14.
Try these: $\mathrm{HCF}(18 \cdot 10, 18 \cdot 15) = 90$; $\mathrm{HCF}(10 \cdot 38, 10 \cdot 21) = 10$; $\mathrm{HCF}(5 \cdot 13, 5 \cdot 20) = 5$; $\mathrm{HCF}(12 \cdot 16, 12 \cdot 20) = 48$.
Common-factor division method: $\mathrm{HCF}(300, 150) = 150$ and LCM= 300; $\mathrm{HCF}(630, 770) = 70$, LCM= 6930. Dividing out composite common factors works too.
Other examples: (90,150) HCF 30, LCM 450; (84,132) HCF 12, LCM 924.

Relation between HCF, LCM and product

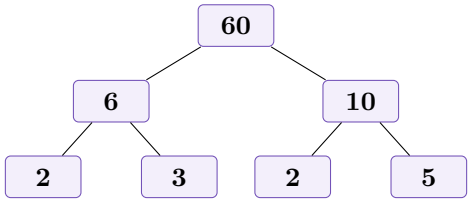
$$\boxed{\mathrm{HCF}(a, b) \times \mathrm{LCM}(a, b) = ab} \qquad (a, b > 0).$$

For (45,105): HCF 15, LCM 315; product $15 \times 315 = 4725 = 45 \times 105$.

For (275,352): HCF 11, LCM 8800; for (222,370): HCF 74, LCM 1110.
Why it works: For each prime, HCF takes the smaller exponent and LCM the larger. Adding those exponents gives the sum of the two original exponents.
Warning: This product identity is for *two* positive integers, not generally three or more.

Part C • Make the mathematics visual

1. Prime-factor tree: break a number down to primes.



$$\boxed{60 = 2^2 \times 3 \times 5}$$

2. The factor ladder: compare the powers of shared primes.

Number	Power of 2	Power of 3	Power of 5	Value
$60 = 2^2 \cdot 3 \cdot 5$	2	1	1	60
$72 = 2^3 \cdot 3^2$	3	2	0	72
HCF: minimum	2	1	0	12
LCM: maximum	3	2	1	360

When should I use HCF or LCM?

HCF: largest identical size for *splitting* or packing with no leftovers.
LCM: earliest common time/position when *repeating* events meet.
A factor divides a number; a multiple is obtained by multiplying the number.

3. Repeating blue stars: the picture-based question.

Top: repeats every 6. Bottom: repeats every 4.



The blue stars meet at 4, then 16: $4 + \text{LCM}(6, 4) = 16$. The first meeting position is an *offset*; LCM gives the *repeat gap*.

Part D • Remember, avoid mistakes, practise

- **1 is not prime.** Its only positive factor is 1, not two different factors.
- For HCF, take *minimum* powers; for LCM, take *maximum* powers.
- Two numbers can be co-prime without either being prime, e.g., 8 and 15.
- HCF can be a number itself if it divides the other: $\text{HCF}(9, 45) = 9$.
- **Check with multiplication:** HCF×LCM must equal product of two numbers.
- Do not use the HCF–LCM product rule for three numbers without checking.

Five one-minute checks • quick answers

1. HCF$(18, 30) = 6$; **2.** LCM$(9, 12) = 36$; **3.** largest square tile $(20, 28)$ cm: 4 cm; **4.** bells every 8 and 12 min: together after 24 min; **5.** HCF= 4, product= 240: LCM= 60.

Extra exam-style practice • with answers

A. Eight quick problems • try before revealing answers

1. HCF$(72, 120)$? 24*: both contain* $2^3 \cdot 3$.
2. LCM$(72, 120)$? 360*: take* $2^3 \cdot 3^2 \cdot 5$.
3. HCF$(28, 45)$? 1*: these are co-prime.*
4. LCM$(28, 45)$? 1260*: their product because HCF is 1.*
5. Buses arrive every 6, 8 and 12 min. Next together? 24 *min.*
6. Two co-prime factors with product 91? 7 *and* 13.
7. If HCF$(a, b) = 6$ and $ab = 756$, LCM? 126.
8. HCF$(84, 126)$ and LCM$(84, 126)$? 42 *and* 252.

B. Three reasoning checks

Why can LCM not be smaller than both positive numbers? It must be a multiple of each, so it is at least as large as the larger one.

Why does HCF$(8, 15) = 1$ not imply 8 and 15 are prime? Co-prime describes the pair, not the primality of either number.

How can you catch a wrong HCF–LCM calculation? Multiply your proposed HCF and LCM. For a pair of positive integers, their product must equal the product of the original numbers.

Remember: HCF splits • LCM synchronises.